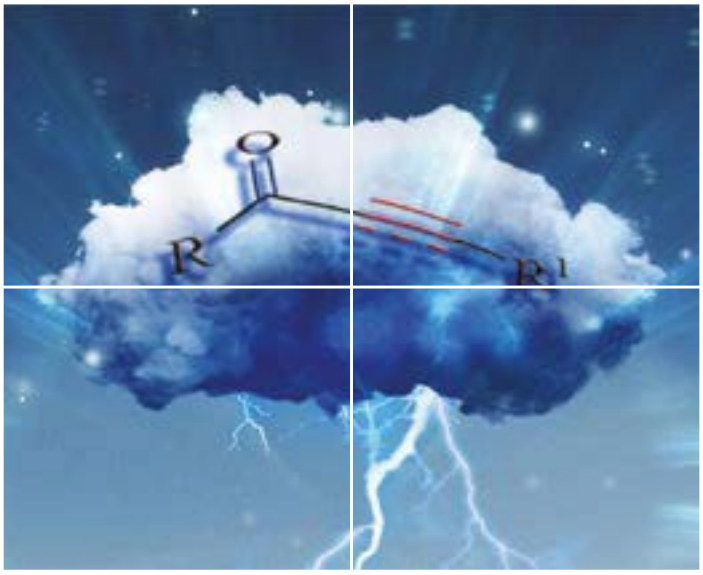
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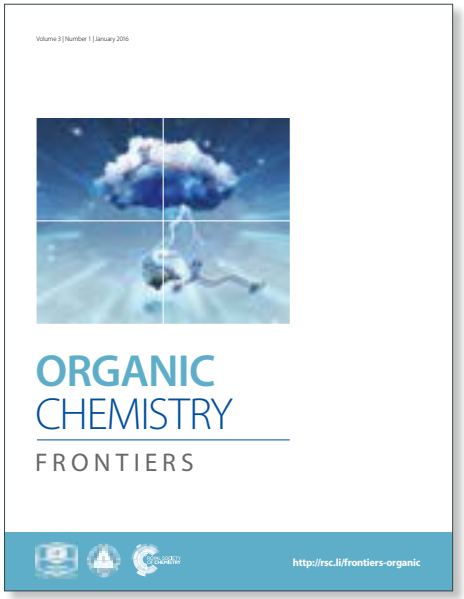
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Asymmetric S_N2'-Type C-H Functionalization of Arenes with Propargylic Alcohols

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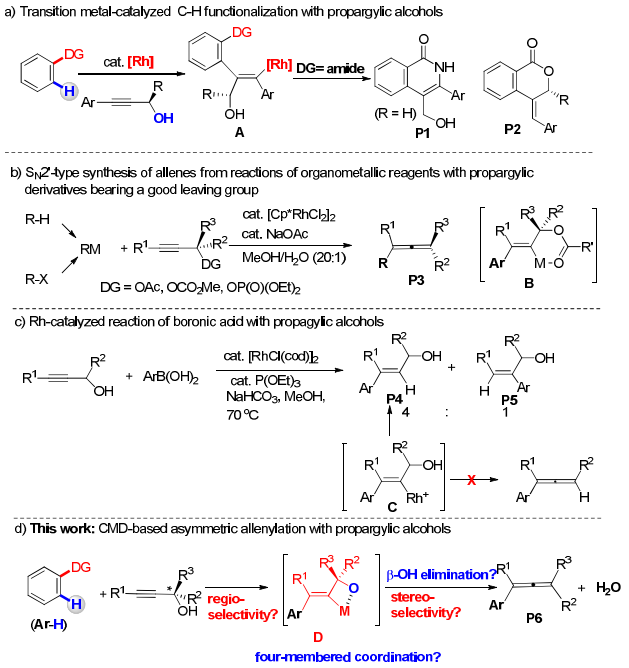
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Shangze Wu, Xin Huang, Chunling Fu, and Shengming Ma*

Herein we present the first example of transition metal-mediated stereodefined β-OH elimination-an atom economic $[Cp^*RhCl_2]_2$-catalyzed S_N2'-type coupling between easily available arenes and non-terminal *tertiary* 2-alkynylic alcohols at rt affording fully substituted allenes with H_2O as the only by-product. When optically active propargylic alcohols were used, asymmetric allenylation was realized with prefect chirality transfer. It is unveiled that the reversed regioselectivity for the C-C triple bond insertion is induced by the steric effect of tertiary carbon center as well as the weak coordination nature of the hydroxyl oxygen with Rh, which is also responsible for the stereospecificity of the rather uncommon β-OH elimination forming allenes.

As we know, C-H functionalization has been become a part of next generation technology for highly efficient synthesis of important functional molecules.[1] On the other hand, alkynes have been used to react with such in-situ generated organometallic species, usually affording a mixture of regioisomeric products.[2] Along this line, propargylic alcohols are a class of common and versatile starting materials in organic synthesis.[3] It would be of high interest and immediate potential if propargylic alcohols may be applied to react with these in-situ generated C-M species since there are a carbon-carbon tripe bond and a central chirality inside this class of molecules. In fact, such reactions with propargylic alcohols have been sporadically studied affording cyclization products **P1** and **P2** with the regioselectivity for the C-C triple bond *syn*-insertion via **A**-type intermediates (Scheme 1a).[2b,2f,2h] Transition metal-catalyzed carbometallation reactions of Grignard reagents with propargylic alcohols also afforded *anti*-carbometallation-based allylic alcohols with the same regioselectivity.[4] On the other hand, transition metal-catalyzed S_N2'-type reactions of pre-formed[5] or CMD-based in-situ generated[6] organometallic reagents with propargylic derivatives with a decent leaving group have been used for the synthesis of allenes[7] although racemization was observed in some cases.[5a,5d-h,5j-l] We envisioned that if the regioselectivity of the C-C triple bond *syn*-insertion could be REVERSED to form **D**-type intermediate, allenes may be formed via subsequent β-OH elimination. In previous studies with carbonates, the most favored six-member-based coordination (intermediate **B**) has been applied to ensure the regio- and stereo-selectivity.[5l,6] However, here, if the weak coordination ability of the hydroxyl group is proposed to work for such a purpose, a challenging four-membered intermediate **D** would be expected. In addition, β-OH elimination with Rh is challenging since Murakami et al. reported that the $[RhCl(COD)]_2$-catalyzed reaction of phenyl boronic acid with propargylic alcohols yielded a mixture of isomeric allylic alcohol products **P4** and **P5**.[8] *No β–OH elimination of Int **C** forming allene has been observed.* In addition, for the purpose of highly stereoselective asymmetric synthesis of chiral allenes **P6** from optically active



Scheme 1. Transition metal-catalyzed carbometalations of propargylic alcohols and their derivatives bearing a leaving group

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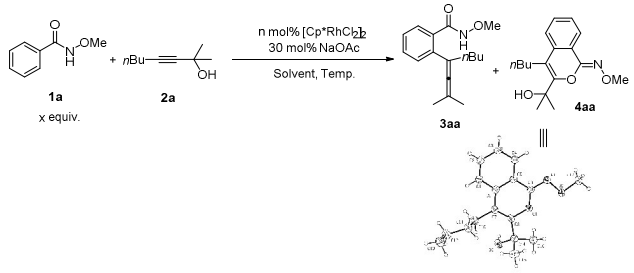
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propargylic alcohols, the stereochemistry of β-OH elimination has to be controlled avoiding commonly observed mixed *syn*- and *anti*-β-OH elimination with Pd.[8,9] During the submission of our manuscript, Sundararaju reported the Cp*Co(III)-catalyzed C-H bond allenylation from propargylic alcohols via β-OH elimination without any chirality transfer example.[10] Herein, we wish to report our own obervation on such a new concept for the asymmetric synthesis of fully substituted allenes via $[Cp^*RhCl_2]_2$-catalyzed[11] coupling between easily available arenes and tertiary propargylic alcohols with an excellent efficiency for point-to-axial chirality transfer and water as the only by-product (Scheme 1d).

Table 1. Optimization of reaction conditions: the Rh-catalyzed coupling of **1a** with **2a.**[a]

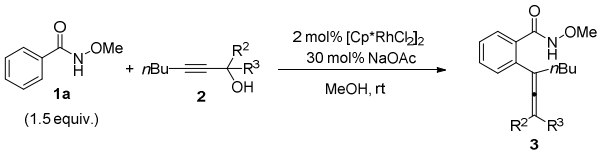


Entry	x	$[Rh]_2$	MeOH/H_2O (v/v)	Time/ h	NMR yield/ %	
					3aa	4aa
1	1.5	4	20/1	24	40	28
2	1	4	20/1	48	32	23
3	1.2	4	20/1	48	41	19
4	1.5	2	20/1	30	60	12
5[b,c]	1.5	4	20/1	72	27	11
6	2	4	20/1	40	48	11
7[d,e]	1	2	20/1	72	58	6
8	1.5	2	10/1	24	55	17
9	1.5	2	5/1	24	50	21
10	1.5	2	1/1	24	47	21
11	1.5	2	100/0	24	65	-

[a] The reaction was conducted with **1a** (entry 2, 0.2 mmol; entry 3, 0.24 mmol; entries 1, 4-5, and 8-11, 0.3 mmol; entry 6, 0.4 mmol), **2a** (0.2 mmol), $[Cp^*RhCl_2]_2$ (entries 4 and 8-11, 0.004 mmol; entries 1-3 and 5-6, 0.008 mmol), NaOAc (0.06 mmol), MeOH (1.2 mL), H_2O (entries 1-6, 0.06 mL; entry 8, 0.12 mL; entry 9, 0.24 mL; entry 10, 1.2 mL) and monitored by TLC. [b] 15% of **2a** was recovered. [c] The reaction was conducted at 0 °C. [d] The reaction was conducted with **1a** (0.5 mmol), **2a** (0.5 mmol), $[Cp^*RhCl_2]_2$ (0.01 mmol), NaOAc (0.15 mmol), MeOH (3 mL), H_2O (0.15 mL). [e] 24% of **2a** was recovered.

To test our hypothesis, readily available *N*-methoxybenzamide **1a** and 2-methyloct-3-yn-2-ol **2a** were used as the model substrates. After some extensive screening it was observed that when **1a** reacted with **2a** in a mixed solvent of MeOH and water (20/1) under the catalysis of $[Cp^*RhCl_2]_2$ at rt,[6,8] based on X-ray diffraction study[12] (entries 1-7, Table 1), a 6-membered oxacyclic product **4aa** was formed in 28% NMR yield.[2b,2f,2h] In addition to that the tetra-substituted allene **3aa** was also formed in 40% NMR yield! Thus, such *β*-elimination is observed for the first time.[8] Based on the structures of both products, the regioselectivity for C-C triple bond insertion is reversed (compare with Scheme 1a). When the reaction was conducted at 0 °C, the yield dropped (Table 1, entry 5). The amount of **1a** could affect the yield and 1.5 equiv. of **1a** were found to be optimal (Table 1, entries 2

Table 2. The scope of the reaction with alkyl-substituted alcohols[a]



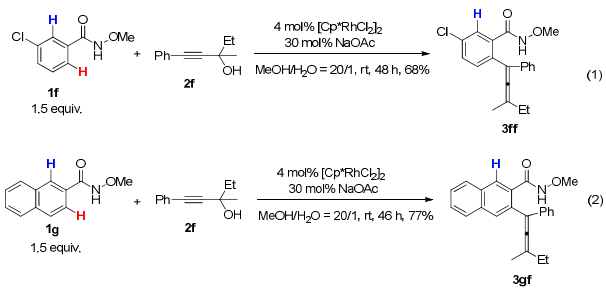
Entry	2 R^2/R^3	Time/ h	Yield of **3**, %
1	Me/Me (**2a**)	34	71 (**3aa**)
2[b]	Me/Me (**2a**)	40	73 (**3aa**)
3[c]	-$(CH_2)_6$- (**2b**)	72	70 (**3ab**)
4	Me/C_5H_{11} (**2c**)	72	66 (**3ac**)
5	Et/Et (**2d**)	36	73 (**3ad**)
6	Me/*p*-$MeOC_6H_4(CH_2)_2$ (**2e**)	60	60 (**3ae**)

[a] The reaction was conducted with **1** (1.5 mmol), **2** (1 mmol), $[Cp^*RhCl_2]_2$ (0.02 mmol), NaOAc (0.3 mmol), and MeOH (6 mL) [b] Reaction was conducted on 8 mmol scale. [c] H_2O (0.3 mL) was added.

and 4-6). Lowering the catalyst loading contributed to a better **3aa**/**4aa** selectivity (Table 1, entries 2 and 7). Finally, the solvent was screened again and when MeOH was used as the solvent, **3aa** was formed in 65% NMR yield, exclusively (Table 1, entry 11).Thus, the reaction of propargylic alcohol **2a** with 1.5 equiv. of arene **1a** in MeOH under the catalysis of $[Cp^*RhCl_2]_2$ (2 mol%) with 30 mol% NaOAc as base (based on further screening of the base effect. See Table S1 in SI for details.) was defined as the optimized reaction conditions for further study (entry 11, Table 1).

Having the optimized reaction conditions in hand, we started to investigate the generality of the reaction (Tables 2 and 3). The R^1 groups in the propargylic alcohol could be alkyl or aryl groups, and R^2 and R^3 could be 1° or 2° alkyl groups (Tables 2 and 3). When R^1 was an aryl group, the reaction was quite slow under the standard conditions. A slightly modification of the condition-using a higher loading of the Rh catalyst and addition of water afforded allenes smoothly (Table 3). The Ar group in **2** may be electron-donating and electron-withdrawing group substituted aryl groups (Table 3, entries 5-7). On the other hand, it was observed that electron-withdrawing synthetically attractive groups such as Br and CO_2Me as well as electron-donating groups such as methoxy and *tert*-butyl in the aryl unit were all well tolerated in the aryl amides **1** giving the corresponding fully substituted allenes in moderate yields (Table 3, entries 8-11). The reaction proceeded well when scaled up, affording **3aa** in 73% yield on a gram-scale (Table 2, entry 2).

It was noteworthy that the less hindered C-H bond in **1f** with 3-Cl was exclusively functionalized forming **3ff** in 68% yield, exclusively (eq 1). When *N*-methoxy 2-naphthyl amide **1g** was



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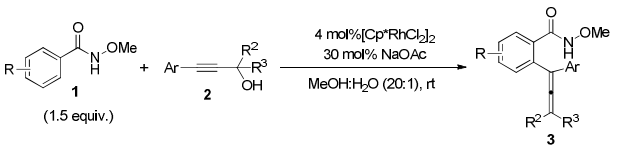
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used, also the less hindered C-H bond was exclusively allenylated giving **3gf** in 77% yield (eq 2).

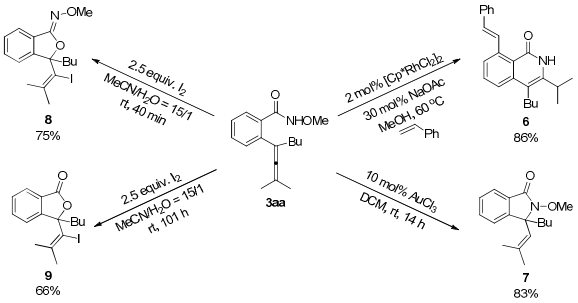
Table 3. The scope of the reaction with aryl-substituted alcohols[a]



Entry	1/2 R/Ar/R^2/R^3	Time/ h	Yield of 3, %
1	H (**1a**)/Ph/Me/Et (**2f**)	72	27[b] (**3af**)
2	H (**1a**)/Ph/Me/Et (**2f**)	34	74 (**3af**)
3	H (**1a**)/Ph/Me/Me (**2g**)	37	70 (**3ag**)
4	H (**1a**)/Ph/Me/Bu (**2h**)	36	72 (**3ah**)
5	H (**1a**)/*p*-$MeOC_6H_4$/Me/Et (**2i**)	72	70 (**3ai**)
6	H (**1a**)/*p*-ClC_6H_4/Me/Et (**2j**)	84	66 (**3aj**)
7	H (**1a**)/*p*-NCC_6H_4/Me/C_5H_{11} (**2k**)	96	61 (**3ak**)
8	4-Br (**1b**)/Ph/Me/Et (**2f**)	60	75 (**3bf**)
9	4-MeO_2C (**1c**)/Ph/Me/Et (**2f**)	72	64 (**3cf**)
10	4-MeO (**1d**)/Ph/Me/Et (**2f**)	42	74 (**3df**)
11	4-*t*-Bu (**1e**)/Ph/Me/Et (**2f**)	45	71 (**3ef**)

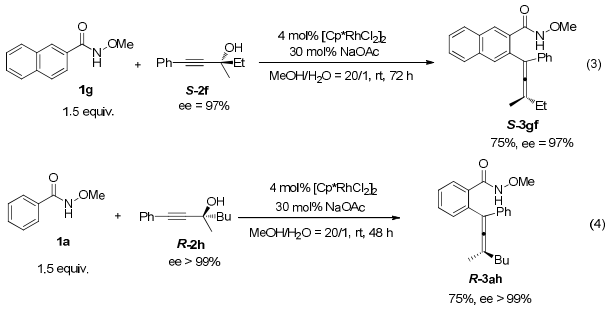
[a] The reaction was conducted with **1** (1.5 mmol), **2** (1 mmol), $[Cp^*RhCl_2]_2$ (0.04 mmol), NaOAc (0.3 mmol), MeOH (6 mL), and H_2O (0.3 mL) and monitored by TLC. [b] The reaction was carried out under the standard conditions for Table 2, **3af** was formed in 27% NMR yield with 65% **2f** recovered.

The formed fully substituted allenes **3** are synthetically quite useful: When reacted with styrene under the catalysis of $[Cp^*RhCl_2]_2$, a C-H activation-styrenylation/allene-cyclization occurred affording *o*-alkenyl substituted benzo-fused six-membered **6**.[13] When $AuCl_3$ was used as the catalyst, benzo fused five-membered cyclization product **7** was formed.[14] If **3aa** was subjected to a I_2-mediated electrophilic cyclization, **8** or **9** was formed depending on the reaction time (Scheme 2).[15] The structures of **6** and **8** were further confirmed by X-ray single crystal diffraction study (See SI for details).[16]



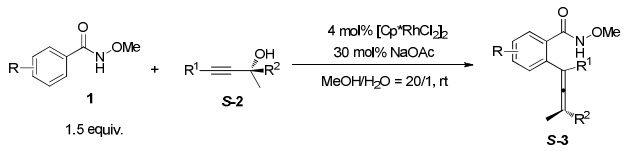
Scheme 2. Synthetic applications

Now with this protocol in hand, we turned to check the stereoselectivity for the β-OH elimination reaction (Table 4 and eqs. 3 and 4), which is the key for the asymmetric synthesis of optically active allenes.[6,17] To our delight, when chiral propargylic alcohol ***S*-2f** was applied, corresponding axial chiral allene ***S*-3af** was afforded with an excellent ee, indicating a highly efficient chirality transfer process (Table 4, entry 1). The absolute configuration of ***S*-3af**[6] confirmed an exclusive *syn*-OH elimination, which strongly supports that the weak coordination between Rh(III) and OH most likely forming a four-membered ring accounts for the highly efficient chirality transfer by stereo-specific *syn*-β-elimination.[8,9] Other amides **1b**, **1d**, **1e**, **1f**, and **1g** as well as other chiral propargylic alcohols ***S*-2h**, ***S*-2j**, ***S*-2i**, and ***R*-2h** could be used to produce the corresponding axially chiral allenes with excellent ee values (Table 4, entries 2-9 and eqs. 3 and 4).



Furthermore, control experiments showed that there is a huge difference between the reaction of **1a** with non-terminal 2-alkynylic alcohol **2l,** which afforded allene **3al** in good yields at either rt (75%) or 50 °C (76%, NMR), and that of terminal alcohol **2m**, which only afforded a trace amount of allene **3am** and the [4+2] product **5am** (at rt or 50 °C) with the same reversed regioselectivity (Scheme 3).

Table 4. Highly enantioselective synthesis of tetra-substituted *S*-allenes by chirality transfer with chiral *S*-propargylic alcohols.[a]



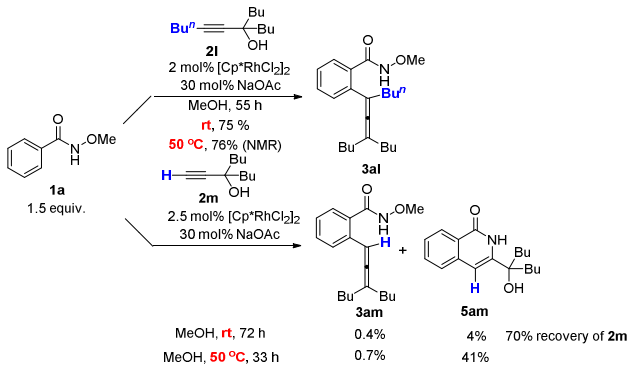
Entry	1/*S*-2 R/R^1/R^2/ee (%)	Time/ h	*S*-3 Yield / %	ee /
1	H (**1a**)/Ph/Et (***S*-2f**)/97	72	70 (***S*-3af**)	97
2	4-Br (**1b**)/Ph/Et (***S*-2f**)/97	72	70 (***S*-3bf**)	96
3	4-OMe (**1d**)/Ph/Et (***S*-2f**)/97	72	68 (***S*-3df**)	97
4	4-*t*-Bu (**1e**)/Ph/Et (***S*-2f**)/97	85	71 (***S*-3ef**)	97
5	3-Cl (**1f**)/Ph/Et (***S*-2f**)/97	72	67 (***S*-3ff**)	94
6	H (**1a**)/Ph/Bu (***S*-2h**)/95	48	73 (***S*-3ah**)	95
7	H (**1a**)/*p*-ClC_6H_4/Et (***S*-2j**)/98	72	63 (***S*-3aj**)	98
8	H (**1a**)/*p*-$OMeC_6H_4$/Et (***S*-2i**)/98	72	74 (***S*-3ai**)	98

[a] The reaction was conducted with **1** (0.45 mmol), **2** (0.3 mmol), $[Cp^*RhCl_2]_2$ (0.012 mmol), NaOAc (0.09 mmol), MeOH (2 mL) and H_2O (0.1 mL) and monitored by TLC. The ee values were determined by HPLC.

In order to know more about the mechanism, (3,3-dimethylbut-1-yn-1-yl)benzene **10** *without a hydroxyl group* was treated with **1a**. Product **11** was formed exclusively via **E-type** intermediate: the regioisomer **11'** from **F-type** intermediate was NOT formed.[18] We reasoned that the steric interaction between bulky *t*-Bu and the H atom in the aryl group stopped the formation of intermediate **F** (Scheme 4a). To further support this hypothesis, secondary alcohol **12** was used to react with **1a**. It was not surprising that **F-type** intermediate was formed leading to the formation of the related cyclization product **13** (Scheme 4b).[2b,2f,2h] These results clearly tell us that the steric effect of the hydroxylalkyl group helps to determine the regioselectivity of the *syn*-insertion of C-C triple bond. However, the slow reaction of **1a**

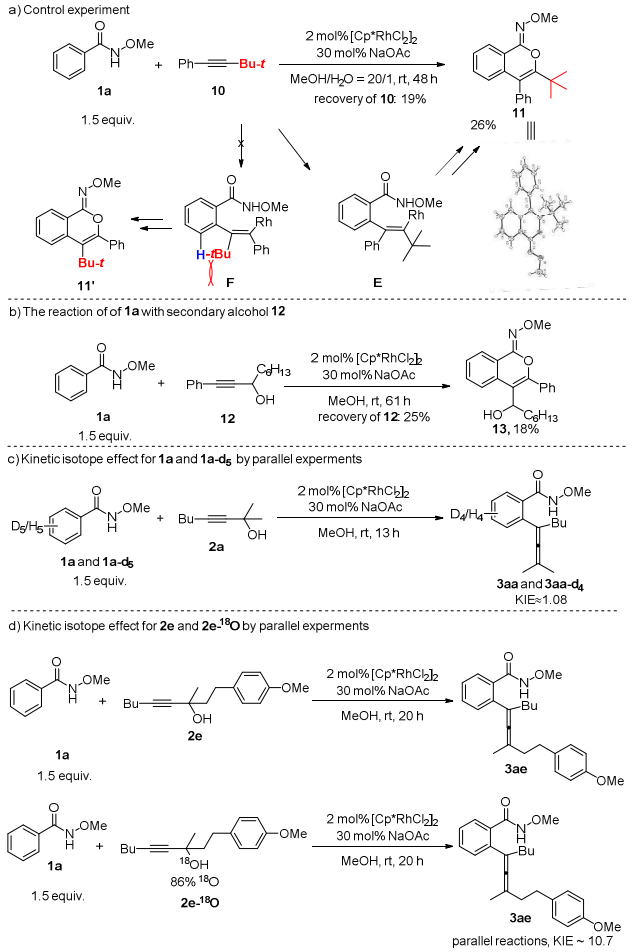
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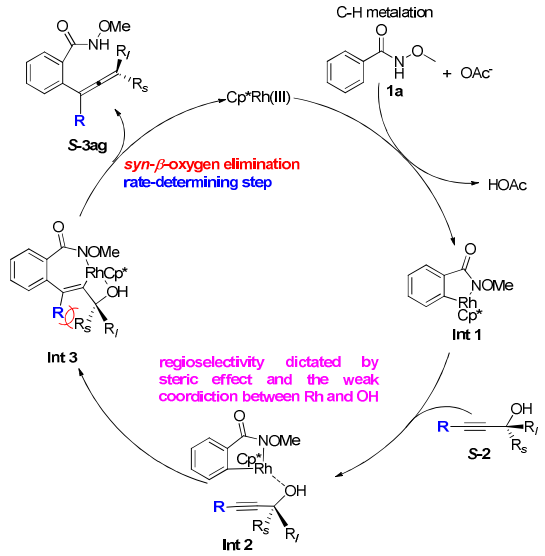
Scheme 3. The difference between terminal and *non*-terminal *tertiary* 2-alkynols

with **10** (Scheme 4a) also supports the importance of the hydroxyl group for the whole reaction in this study. Thus, the Thorpe-Ingold effect[19] of in the *tertiary* alcohol unit may push the hydroxyl group for its easier coordination with Rh. Finally, the kinetic isotopic effects (KIE) of **1a/1a-d$_5$** and **2e/2e-^{18}O** were measured by parallel experiments: A small KIE value of about 1.0 of **1a/1a-d$_5$** indicated that C-H activation should not be the rate-determining step (Scheme 4c).[20] By comparison, the KIE value of **2e/2e-^{18}O** was ~ 10.7, showing that the β-OH elimination must be the rate-determining step (Scheme 4d).



Scheme 4. Mechanistic studies

Based on these results, a plausible mechanism has been shown in Scheme 5: The first step of the C-H activation of arenes would form rhodacyclic intermediate **Int 1**, which is followed by the coordination with the C-C triple bonds in ***S*-2** affording **Int 2.** Subsequently, steric effect of the tertiary carbon group and the weak coordination ability of the hydroxyl group dictated the reversed regiospecific *syn*-insertion giving the hydroxyl oxygen-coordinated four-membered rhodacyclic intermediate **Int 3**. **Int 3** underwent *syn*-β-OH elimination to form the final product ***S*-3** and regenerates the catalytically active Rh(III) species**.** As a comparison, it should be noted that the palladium-catalyzed coupling reactions of aryl boronic acids with non-terminal propargylic alcohols led to the formation a mixture of allenes and alkynes while the reaction with terminal 1-phenyl propargyl alcohol (92% ee) afforded allene in only 10% ee.[21] The results in Scheme 3 clearly showed that the R group in **Int 3** may promote the β-OH elimination by steric effect.



Scheme 5. A proposed mechanism.

Conclusions

In conclusion, we have reported the first example of transition metal-catalyzed *syn*-β-OH elimination reaction, which led to a new protocol for the efficient and asymmetric synthesis of not-readily available tetra-substituted chiral allenes using easily available arenes and non-terminal 2-alkynylic alcohols as the starting materials at rt with decent yields. The reaction is compatible with air, moisture and many synthetically useful functional groups. The regioselectivity for alkyne insertion has been induced by the Thorpe-Ingold effect of tertiary carbon unit and the weak coordination ability of the hydroxyl group with Rh, which also accounts for the exclusive *syn*-β-OH elimination. Considering the easy availability of both starting materials and the prefect efficiency of chirality transfer as well as the versatile ways to generate species containing C-M bonds including C-H activation and transmetalation, this protocol will open new approaches to chiral allenes. Further studies in this area are being pursued in our laboratory.

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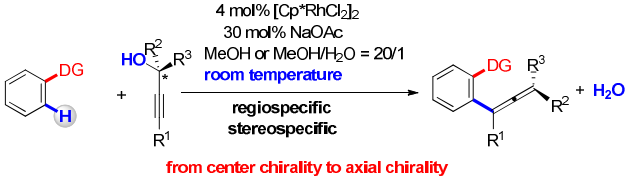
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The first example of transition metal-catalyzed stereo-defined *syn*-β-OH elimination! A $[Cp^*RhCl_2]_2$-catalyzed allenylation reaction of available arenes with propargylic alcohols afforded fully substituted allenes with excellent ees.

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